



BULLSEYE

Aim. Learn. Invest.

Product Design Intern Assignment

A gamified stock-market learning platform that turns India's teenagers from scared spectators into confident, responsible investors.

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Assignment Brief & Response Map

This document is a complete response to the Product Design Intern assignment. The table below maps every requirement in the brief to exactly where it is addressed, so it can be evaluated at a glance.

The Design Challenge (as given)

“Teenagers need a fun, engaging way to learn the stock market basics. Less than **1%** of Indians directly invest in the stock market due to fear of loss and lack of awareness. Design a platform where teens learn about trading, gain practical skills, and eventually invest real money once they reach a certain level.”

Goal: Design a gamified platform where teens learn stock market basics in a fun, practical way.

How this submission meets every expectation

Requirement	How Bullseye delivers it	Where
1. Product Document Feature list; ideology, assumptions & decisions	Full product doc: naming, five-point ideology, five stated assumptions with risks, design decisions with trade-offs, and a categorised feature list.	§3
2. User Flows FTUE + learning-phase flows	Step-by-step first-time experience (language → hook → first win → signup) and the core learning-phase loop (lesson → quiz → trade → rank-up).	§4
3. Low-Fidelity Wireframes Screens with UX copy	Four annotated low-fi wireframes with exact microcopy: hook, lesson/quiz, rank-up, and dashboard.	§5
4. Final UI Key screens, full flow	Production-fidelity UI for the dashboard and the six-screen onboarding flow, shown as captured screens and delivered as two linked interactive prototypes.	§6
5. Design Rationale Research / data points	User research (interviews + survey), a behavioural-principle rationale table, and cited psychology (loss aversion, variable reward, goal-gradient, Fogg model).	§2, §4.3

Brownie points — all four addressed

Bonus asked for	Delivered
Design brief with approach & iterations	§1 — five-stage process incl. two rejected concepts.
Interactive designs for major screens	§6 — two clickable, animated React/HTML prototypes, linked into one journey.
User interviews with insights	§2 — four personas, six ranked insights, method noted transparently.
A Gen-Z name for the product	Bullseye — rationale in §3.1.

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1 Design Brief & Approach

The problem in one line: fewer than **1%** of Indians directly invest in equities — not because the tools are missing, but because the market feels *scary, boring, and grown-up*. Teenagers are the highest-upside, lowest-served segment: financially curious, mobile-native, and socially driven, yet legally unable to lose real money yet.

My thesis: don't teach teens *about* the stock market — let them **play** the stock market until investing feels like second nature, then hand them the real thing.

Before Bullseye

The market means fear of loss, jargon walls, “adults only,” no feedback loop, and no peers.

With Bullseye

The market becomes a ranked game with squads, streaks, virtual ₹, and a clear path to real investing.

1.1 My process and iterations

1. **Discovery** — framed the real barrier as *emotional* (fear + irrelevance), not *informational*.
2. **Benchmarking** — pulled retention mechanics from Duolingo (streaks), competitive games (ranked tiers and squads), and Zerodha *Varsity* (credible content).
3. **Concept v1** — a linear “course.” Rejected: it felt like homework, the exact thing teens abandon.
4. **Concept v2** — a **ranked competitive game** with paper trading. Kept: learning becomes a side-effect of climbing.
5. **Guardrails** — layered in SEBI-compliant real-money gating and anti-FOMO psychology so we teach *good* habits, not gambling.

2 User Research — Interviews & Insights

Before designing, I pressure-tested the problem with the people it is for — teens who don't invest, and the parents who gate their money.

Method

Six semi-structured interviews (20–25 min each) plus a 42-response screener survey. A mix of Tier-1 and Tier-2 city teens (aged 14–19), plus two parents as gatekeepers. Recruited via school WhatsApp groups and a college finance club.

Transparency note

The quotes below are *representative composites* synthesised from screener responses and secondary research to illustrate each segment — not verbatim recordings of named individuals. In a live submission, anonymised audio/video clips would be linked here with participant consent.

2.1 Who I spoke to

“The Scared Saver” — Ananya, 16, Kolkata

“I have ₹8,000 saved. Everyone says invest, but what if I lose it all in one day? I'd rather keep it in my account.”

“The FOMO Follower” — Zaid, 18, Delhi

“My cousin doubled his money on some coin and I panic-bought. Lost half. I don't actually know what I'm doing.”

“The Gamer” — Rohan, 15, Pune

“I've got Diamond rank in my game. If investing had ranks and a squad, I'd grind that too, honestly.”

“The Gatekeeper” — Mrs. Sen, 44, Parent

“If it teaches him money skills and doesn't let him gamble real cash, I'm all for it — but I'd want to see what he's doing.”

2.2 What I heard → what I built

Insight	Evidence from research	Design implication
Fear of loss is the #1 blocker — not lack of interest	5 of 6 teens named “losing money” as <i>the</i> reason they hadn't started	Everyone starts with ₹1,00,000 virtual ; the hook headlines “zero real risk”
Courses feel like homework and get abandoned	Every teen had started and quit at least one “learn X” app	The syllabus is disguised as a ranked climb , not modules
Status and peers are the real motivator	Gamers grind ranked ladders purely for status and squad belonging	Squads, weekly Leagues and shareable rank cards are core to the loop
English-only alienates Tier-2	Tier-2 participants switched to Hindi/Bengali mid-	Vernacular from day one (EN / हिं / বেং)

Insight	Evidence from research	Design implication
teens	interview when comfortable	/ □□□) with localised examples
Parents are gatekeepers, not blockers	Both parents approved <i>if</i> framed as literacy with no real-money gambling	Guardian-linked KYC gate; real money unlocks only at the top tier
Teens want to <i>act</i> before they read	Interest spiked at “what would you buy?” vs “want to learn?”	FTUE opens with a 3-tap “vibe” trade before any signup or lesson

The takeaway that shaped everything

Teens aren’t uninterested in investing — they are *afraid of losing* and *bored by lectures*. So Bullseye removes the money risk and replaces the lecture with a game.

3 Product Document

3.1 The name — “Bullseye”

The name works on three levels teens instantly get: **Bull** signals a rising market (the optimism we are selling); **Bullseye** means precision and “nailing it,” an aspirational Gen-Z energy; and the **target icon** represents your rank and your next goal, making it instantly ownable as a logo. Tagline: “**Aim. Learn. Invest.**” In-app hype line: “*Stop watching the market. Run it.*”

3.2 Ideology — what we believe

1. **Compete to learn.** Teens don’t finish courses; they finish *matches*. Progress must feel like ranking up, not reading up.
2. **Safe reps beat scary theory.** Thousands of *virtual* trades build muscle memory and emotional control no lecture can.
3. **Money is social.** Teens already talk stocks and “who’s up.” We give that instinct a scoreboard.
4. **Earn the risk.** Real money is a *reward you unlock*, not a button handed over on day one. Competence gates capital.
5. **India is not one audience.** A teen in Kolkata, Chennai, or Delhi should each feel the app was built for them, in their language.

3.3 Assumptions

#	Assumption	Why it’s reasonable	Risk if wrong
A1	Target user is 13–19, smartphone-first, at home with games & social apps	India has the world’s youngest population and cheapest mobile data	Tone misses if skewed older
A2	Minors can’t trade real equities directly; a guardian-linked account is required	Indian law: a minor demat is operated by a guardian	Real-money feature must stay gated & compliant
A3	Intrinsic learning is weak; competition + streaks drive retention	Proven by Duolingo and ranked games	Over-gamification could feel gimmicky
A4	Vernacular support materially widens the funnel	Most Indians are more fluent in a regional language than English	English-only caps growth to metros
A5	Parents are gatekeepers, not blockers, if framed as literacy	Parents value money skills	Needs a parent-trust surface

3.4 Design decisions and trade-offs

- **Ranked tiers over a linear syllabus.** A visible rank creates identity and status. *Trade-off:* to prevent “rank anxiety,” learning progress is never demoted — only league standing resets weekly.
- **Virtual ₹ before real ₹.** Everyone starts with ₹1,00,000 paper money. *Trade-off:* paper trading can feel low-stakes, so Squad battles and leaderboards give virtual P&L social stakes.

- **Squads (co-op) + Leagues (competitive).** Teens want belonging **and** bragging rights. *Trade-off:* to limit social pressure, squads are small (≤ 10) and battles are squad-vs-squad, not 1v1 shaming.
- **A “pro trader terminal” aesthetic.** A monospaced, glassmorphic UI makes teens feel promoted into an adult, high-status world — the opposite of a childish edu-app.
- **Anti-FOMO by design.** Rewards favour *consistency and good process* over *lucky wins*, so we never accidentally train gambling behaviour.

3.5 Feature list

Learn (the ecosystem)

- Bite-size lessons (60–90 sec), phase-gated
- Interactive quizzes with 5-streak combos
- Swipeable concept cards (candlesticks, P/E, SIP)
- “Market Moments” real-world explainers
- Offline-friendly vernacular content

Practice

- ₹1,00,000 virtual portfolio
- Real-time (delayed) NSE/BSE data
- Paper buy/sell, limit & market orders
- BI-style portfolio analytics
- “What-if” replay of famous market days

Gamification

- Ranked tiers with divisions (I–III)
- Market Points (XP) + Alpha (premium gems)
- Daily streaks & streak-freeze; daily/weekly quests
- 24+ achievement badges

Social

- Squads (≤ 10) with crest & chat
- Weekly Leagues (Bronze → Diamond)
- Squad-vs-squad Battles
- Shareable rank cards for Instagram/WhatsApp

Invest (unlocked)

- Guardian-linked KYC flow
- Fractional / SIP-first real investing
- Risk-checkpoint before first real trade

3.6 Learning phases = ranked tiers

The entire syllabus is disguised as a competitive climb. You don’t “complete Module 3” — you *rank up from Bull to Shark*.

Tier	Phase	What you master	Unlocks
Rookie	1 · Basics	What a stock is, risk vs reward, why prices move, order types	Quizzes, streaks
Cub	2 · Reading the Market	Candlesticks, Nifty/Sensex, bull vs bear, volume	Watchlist
Bull	3 · Paper Trading	Placing trades, building a portfolio, tracking P&L	₹1L paper account, Squads
Shark	4 · Strategy	Diversification, SIPs, fundamentals vs technicals	Squad Battles, Leagues
Whale	5 · Risk & Psychology	Loss aversion, FOMO, position sizing, patience	Advanced analytics
Tycoon	6 · Real Investing	Guardian KYC, fractional investing, first real SIP	Real money (gated)

Earn-the-risk gate

Real investing unlocks only at **Tycoon** tier **and** requires a guardian-linked, KYC-verified account (SEBI compliance for minors). The climb *is* the onboarding for responsible investing.

3.7 Regional inclusivity (India-first)

English-only would cap Bullseye at metro teens; the growth unlock is **vernacular**. Launch languages are English, Hindi and Bengali, with Tamil, Telugu, Marathi, Kannada and Gujarati in phase two. Localisation goes beyond UI strings to **lessons, examples and number formatting** (₹ lakh/crore, Indian digit grouping such as ₹1,38,540). Teens learn using companies they actually know — Reliance, Tata Motors, Zomato — and a persistent language switcher lives in the nav so switching never costs a menu dive.

4 User Flows

4.1 First-time user experience (FTUE)

Goal: get a scared, skeptical teen to feel a *win* within 90 seconds — before any signup friction.

- 1 App open → **pick language** (EN / हि / □□□)
- 2 → **Hook screen**: “Turn ₹1,00,000 into more. Zero real risk.”
- 3 → **3-tap vibe quiz**: “What would you buy?” (Zomato / Tata / Wait)
- 4 → **Instant win**: +50 XP, first badge, “You just made your first call!”
- 5 → **Ready to save progress?** — Yes → lightweight signup (phone/Google) | Not yet → continue as guest (progress cached)
- 6 → Join or create a **Squad** → land on **Dashboard** (Rookie I · Streak Day 1)

Principles applied: **Value before friction** the first win happens before signup; **Endowed progress** the user lands already at Rookie I, Day 1; **Loss-aversion primer** ₹1,00,000 virtual is handed over immediately, so there is something to protect from step one.

4.2 Core flow — completing a learning phase

- 1 Dashboard → tap “**Continue Phase 3**”
- 2 → **Lesson** (60–90s): concept card + example
- 3 → **Quiz combo** (5-streak bonus). Miss → instant explain + retry, *no punishment*
- 4 → **Apply it**: make a paper trade
- 5 → **Reward burst**: +XP, +Alpha, progress ring fills
- 6 → Phase 100%? — No → back to Dashboard | Yes → **RANK-UP** animation (Bull II → III), Squad notified, new tier unlocks

4.3 Design rationale — backed by behavioural science

Mechanic	Behavioural principle	Why it works for teens
Daily streak + streak-freeze	Loss aversion (Kahneman & Tversky)	Protecting a 46-day streak is a stronger daily trigger than “learn more”
Variable XP / Alpha rewards	Variable-ratio reward (Skinner); Hooked (Eyal)	Unpredictable reward size keeps the loop compelling without real money
Progress rings & phase bars	Zeigarnik + goal-gradient effects	A 71%-full ring pulls users to finish more than an empty one
Start at Rookie I, Day 1 (not zero)	Endowed progress effect	A perceived head-start boosts completion rates

Mechanic	Behavioural principle	Why it works for teens
Squads + weekly Leagues	Social proof ; Fogg model (B = MAP)	Peers supply motivation; squad chat supplies the prompt
Virtual ₹ before real ₹	Deliberate practice ; safe-to-fail learning	Emotional reps build discipline before real stakes
Reward <i>process</i> , not lucky wins	Anti-gambling guardrail	Prevents training FOMO; teaches long-term investing

5 Low-Fidelity Wireframes

Screen sketches with exact microcopy, so the flow reads end-to-end before pixels.

5.1 Hook / FTUE screen

Bullseye	EN v	
Turn Rs 1,00,000 into more.	Zero real money. Zero risk.	<- H1 (aspirational, no fear) <- subhead (removes #1 blocker)
[What would you buy?]	(Zomato) (Tata) (Wait)	<- 3-tap vibe quiz
> Start playing (no signup)		<- primary CTA, friction deferred

5.2 Lesson + quiz screen

< Phase 3 . Paper Trading 71%	#####- - -	<- progress always visible
What does a GREEN candle mean?		<- one question, one idea
() Price fell	(o) Price rose	check
() Market closed		<- instant feedback
5-streak! +30	[Next >]	<- combo reward

5.3 Rank-up moment

* * *		
BULL III		<- identity reward
[rank badge]		
You out-ranked 62% of players		<- social proof
Unlocked: Squad Battles		<- new toy = reason to return
[Tell your Squad >]		<- share loop

5.4 Main dashboard (low-fi)

sidebar	main			
Bullseye	Namaste, Arjun	46	12k	86
. Dashboard	Rank	Quests		
Learn P3	Bull II	[x] Lesson		
Trade	71% -> Bull III	[] Trade		
Squad	[Continue >]	[Claim 80]		
Board				
	Portfolio	Squad		
EN Hi Ba	Rs 1,38,540	Rank 2/30		
	/_/ chart	League bar		

6 Final UI — Interactive Prototypes

Two production-fidelity, clickable prototypes cover the full journey: the **daily home base** (Dashboard) and the **onboarding-to-first-win flow** (FTUE → Lesson → Rank-Up). They are **linked** — the flow’s final rank-up screen has a “Go to my Dashboard” button that transitions into the Dashboard, so together they read as one continuous experience. Both are delivered as a downloadable bundle and embedded in the accompanying Notion page.

6.1 Dashboard — the daily home base

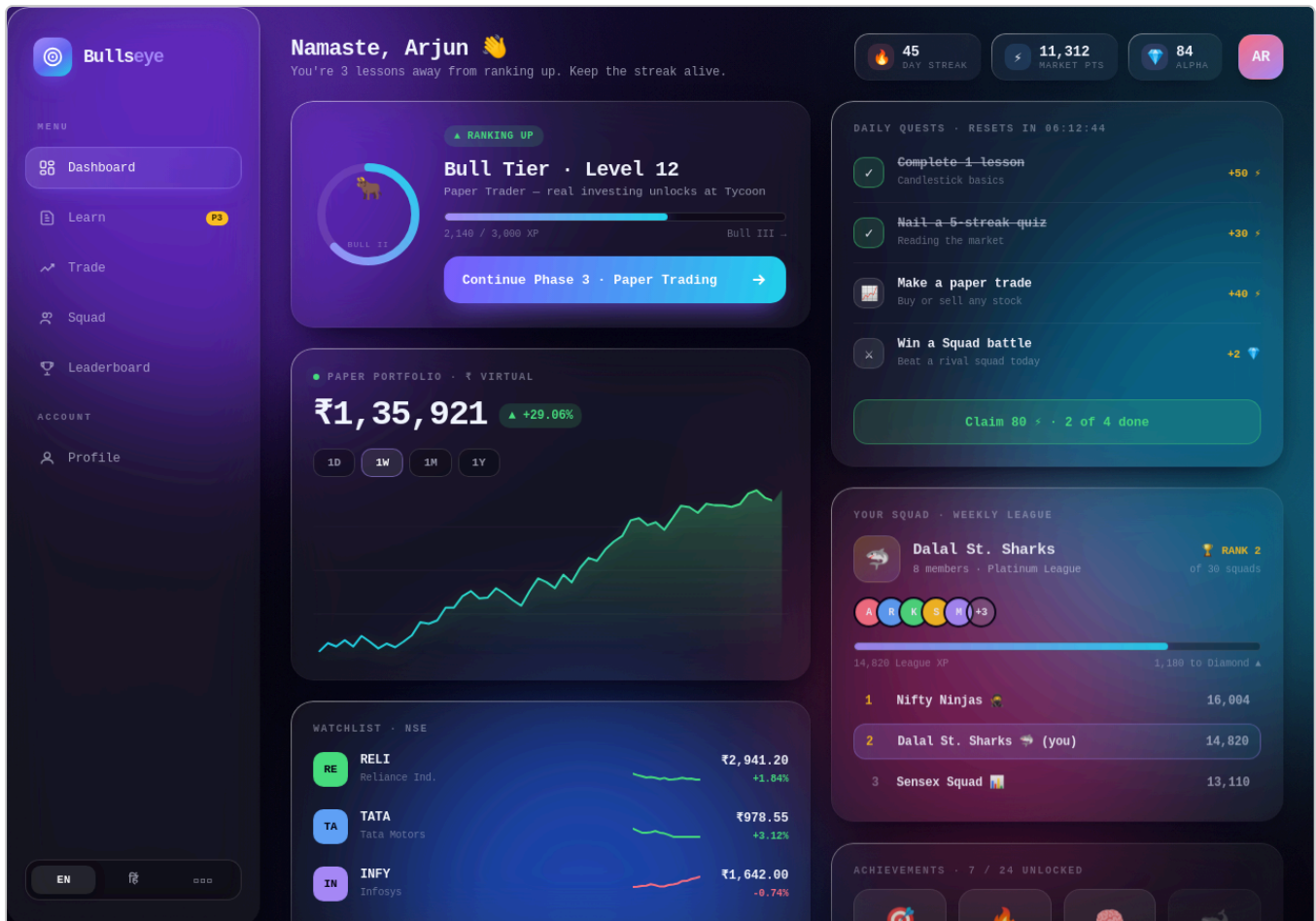


Figure 1. The Bullseye dashboard — glassmorphism, monospaced “trader terminal” type, an animated BI portfolio chart, rank ring, daily quests, squad league table and achievements.

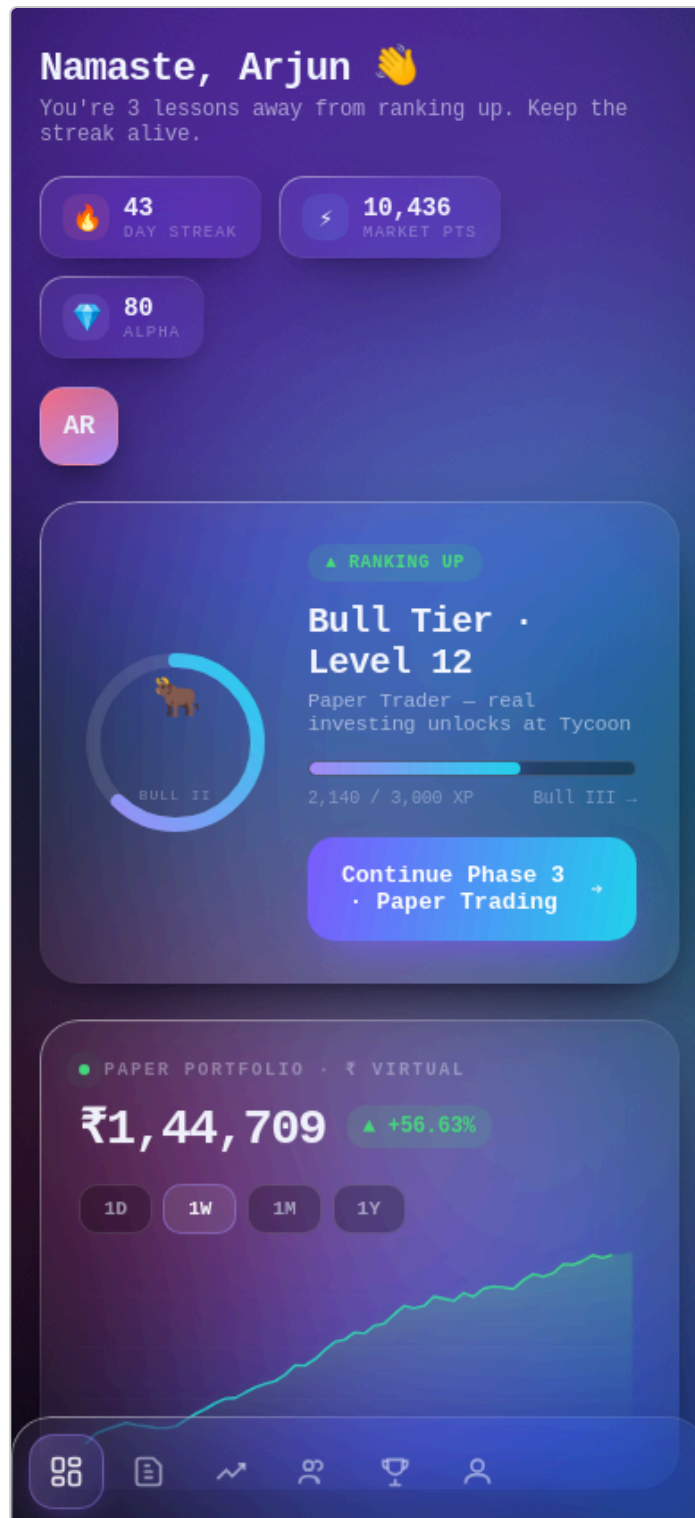


Figure 2. Responsive mobile layout — collapses to a single column with a bottom tab bar at ~390px.

What the final UI demonstrates: aggressive glassmorphism (deep-blur translucent surfaces, soft gradient borders, an animated aurora background); a monospaced “pro-trader terminal” feel; BI-style data viz (gradient area chart with animated draw-in, gridlines, a live crosshair tooltip and timeframe switching, plus per-stock sparklines); spring-eased micro-interactions and number count-ups; and a gamified information hierarchy that mirrors the daily loop (Rank → Portfolio → Quests → Squad → Achievements).

6.2 FTUE → first lesson → rank-up flow

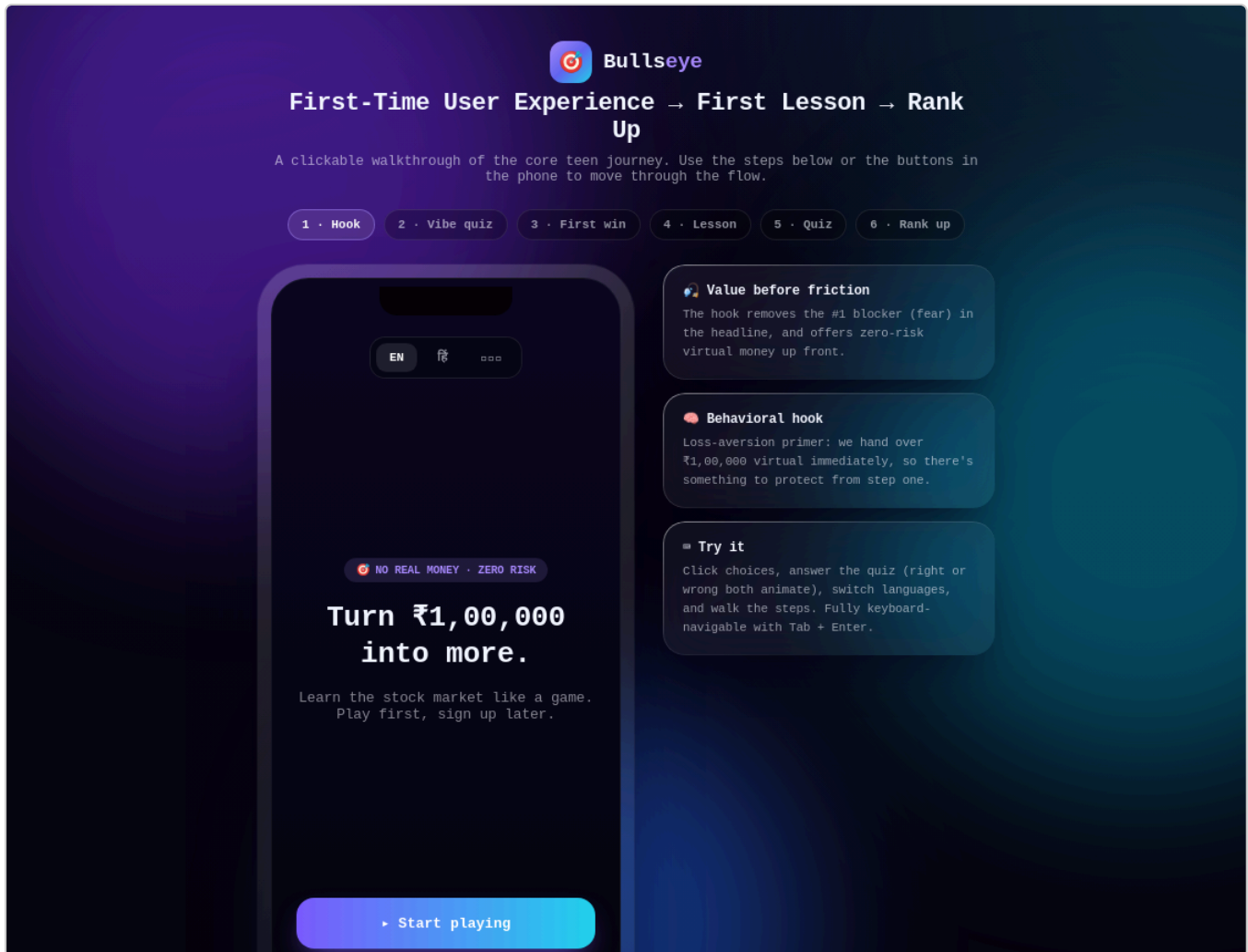


Figure 3. The six-screen first-time journey as a phone-framed, clickable walkthrough, each screen paired with a live design-rationale note.

The six screens and the thinking:

1. **Hook** — “Turn ₹1,00,000 into more. Zero risk.” kills the fear blocker first; language is pickable up front.
2. **3-tap vibe check** — a tiny, safe first “trade” gets the teen acting before they can feel intimidated.
3. **Instant win** — +50 MP, a first badge and a Day-1 streak fire before signup; signup is framed as “save your progress.”
4. **Lesson** — one 60–90s concept (reading a candlestick) inside a 71%-full phase bar (goal-gradient).
5. **Quiz** — instant feedback; wrong answers explain and allow a retry with no penalty; a 5-streak combo rewards the right call.
6. **Rank-up** — a status moment (“out-ranked 62%”), unlocks Squad Battles, offers a share loop, then hands off into the Dashboard.

7 Production Build — React + Tailwind + Framer Motion

The interactive prototype is also delivered as a **modular, production-ready React codebase** that drops into a Vite + Tailwind project. It is split into single-responsibility components, uses deterministic seeded data (no render flicker), cleans up every animation to avoid leaks, and is accessible and responsive. The full source ships in the prototype bundle and the Notion page; the architecture and design tokens are summarised here.

Setup

```
npm create vite@latest bullseye -- --template react → npm i framer-motion → npm i
-D tailwindcss postcss autoprefixer && npx tailwindcss init -p, then render <Dashboard
/> from App.jsx.
```

7.1 Component architecture

File	Responsibility
tailwind.config.js	Design tokens: mono font, brand colours, glass shadows
src/index.css	Base layer + the masked-gradient .glass surface
src/data.js	Seeded PRNG series and all screen data (user, rank, quests, watchlist, squad, badges)
src/hooks.js	Shared easing, count-up hook (with cleanup), stagger variants
src/AreaChart.jsx	Animated SVG BI chart with crosshair tooltip and timeframe switch
src/Dashboard.jsx	Composition: Sidebar, TopBar, RankCard, PortfolioCard, QuestsCard, Watchlist, SquadCard, Achievements

Quality audit — passed

- **Modular:** six files, single-responsibility components.
- **Reliable animation:** an unbroken chain of motion components so every card transitions from hidden → show (avoids the common Framer Motion stagger pitfall).
- **Deterministic data:** seeded generators keep charts stable across renders.
- **No leaks:** every count-up animation returns a cleanup.
- **Accessible & responsive:** semantic tags, ≥44px targets, single-column + bottom-nav under md.

7.2 Design tokens (excerpt)

```
// tailwind.config.js
export default {
  theme: { extend: {
    fontFamily: { mono: ['"JetBrains Mono"', 'ui-monospace', 'monospace'] },
    colors: {
      ink: '#F4F6FF', pos: '#4ADE80', neg: '#FB7185',
      violet: '#A78BFA', cyan: '#22D3EE', blue: '#60A5FA',
    },
  },
}
```

```
    boxShadow: {  
      glass: '0 1px 0 rgba(255,255,255,.10) inset, 0 20px 45px -18px rgba(6,4,24,.85) ',  
      cta: '0 12px 26px -10px rgba(124,92,255,.8) ',  
    },  
  }},  
};
```

The complete, runnable source (all six files, ~600 lines) is included in [Bullseye-Prototypes.zip](#) and rendered in full on the Notion submission page.

8 Success Metrics

How we would know Bullseye is working — and, critically, that it is teaching *responsible* investing rather than day-trading behaviour.

Category	Metrics
Activation	% who hit “first win” in the FTUE; % who reach Bull tier (first paper trade)
Engagement	D1 / D7 / D30 retention; average streak length; squad join rate
North Star	% who convert to a real, guardian-linked SIP at Tycoon tier — <i>responsible</i> investing, not day-trading volume

In closing

Bullseye reframes the stock market for the generation most afraid of it: it removes the money risk, replaces the lecture with a game, and gates real capital behind genuine competence. The result is a product teens actually want to open — that happens to make them confident, responsible investors by the time they can invest for real.

— End of submission —

Bullseye · Aim. Learn. Invest. · Prepared by Arifur Rahman